CMP - Comprehensive Project Map
نظام الحسابات المتكامل | IntegratedAccountsSystem

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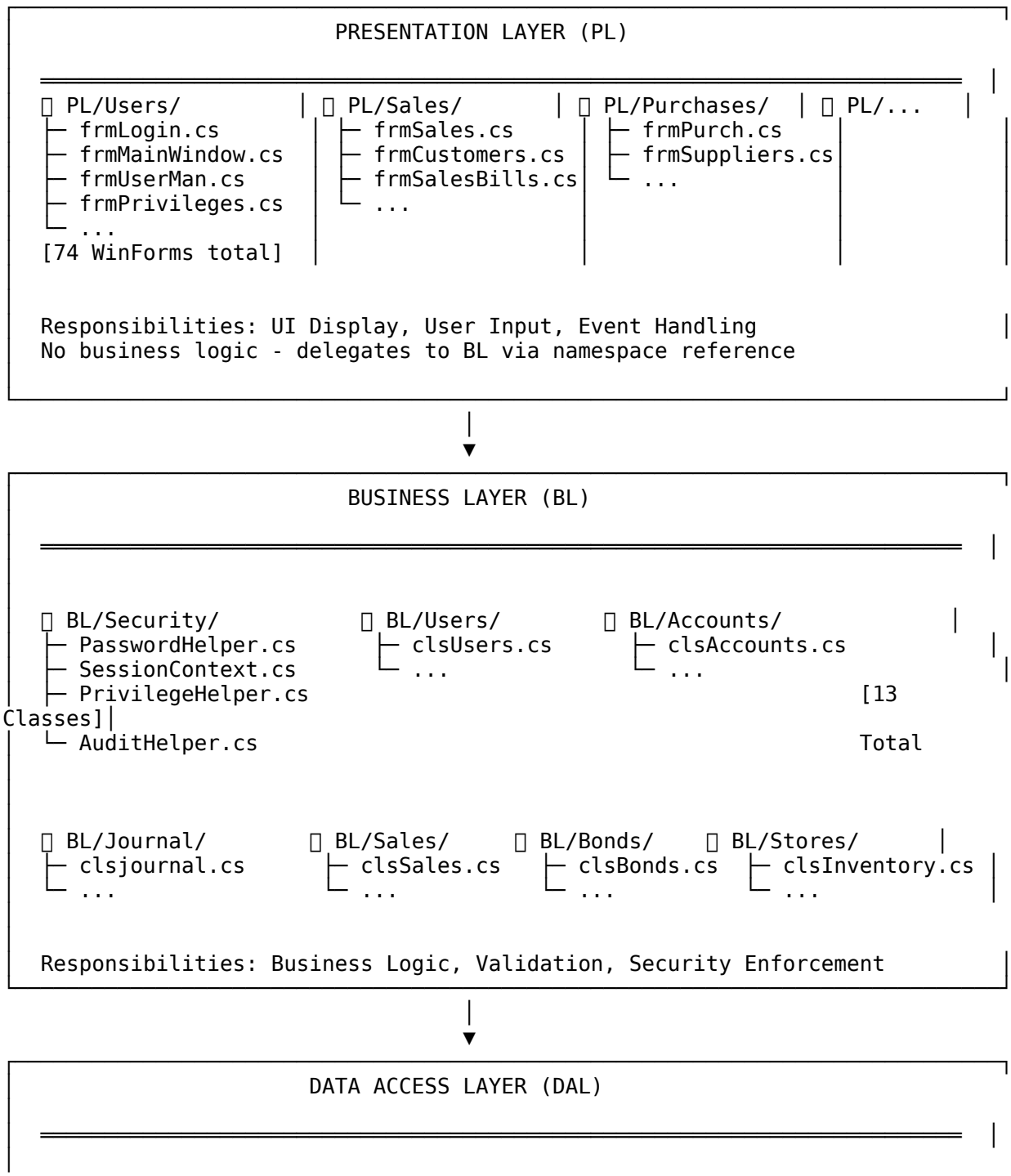
1. البنية المعمارية الكاملة | Full Architecture



```
│                                                                              │
│  📁 DAL/                                                                     │
│  └─ clsCN.cs (187 lines)                                                     │
│     ├─ ExecuteNonQuery()                                                     │
│     ├─ ExecuteReader()                                                       │
│     ├─ GetDataTable()                                                        │
│     │                                                                        │
│     ├─ ValidateStoredProcedureCall() → SQL Injection Prevention              │
│     └─ Transaction Support (BeginTransaction/Commit/Rollback)                │
│                                                                              │
│  Responsibilities: SQL Execution, Connection Management, SP Validation       │
└──────────────────────────────────────────────────────────────────────────────┘
                                    │
                                    ▼
┌──────────────────────────────────────────────────────────────────────────────┐
│                          DATABASE (SQL Server)                               │
│  ═══════════════════════════════════════════════════════════════════════     │
│                                                                              │
│  📁 Database/                                                                │
│  ├─ Stored Procedures (Session Management)                                   │
│  │  ├─ createSession        - Create new session with token                  │
│  │  ├─ validateSession      - Validate token, check expiry                   │
│  │  ├─ updateSessionActivity- Sliding expiry update                          │
│  │  ├─ endSession           - Explicit logout                                │
│  │  └─ expireOldSessions    - Batch cleanup of expired sessions              │
│  │                                                                           │
│  └─ Database Schema (Tables, Views, Functions)                               │
│     ├─ Users, Privileges, UserPrivileges                                     │
│     ├─ ChartOfAccounts, JournalHeader, JournalBody                           │
│     ├─ Sales, Purchases, Inventory, Bonds                                    │
│     └─ Sessions table with Token/Expiry tracking                             │
│                                                                              │
│  Responsibilities: Data Persistence, Integrity, Query Execution              │
└──────────────────────────────────────────────────────────────────────────────┘
```

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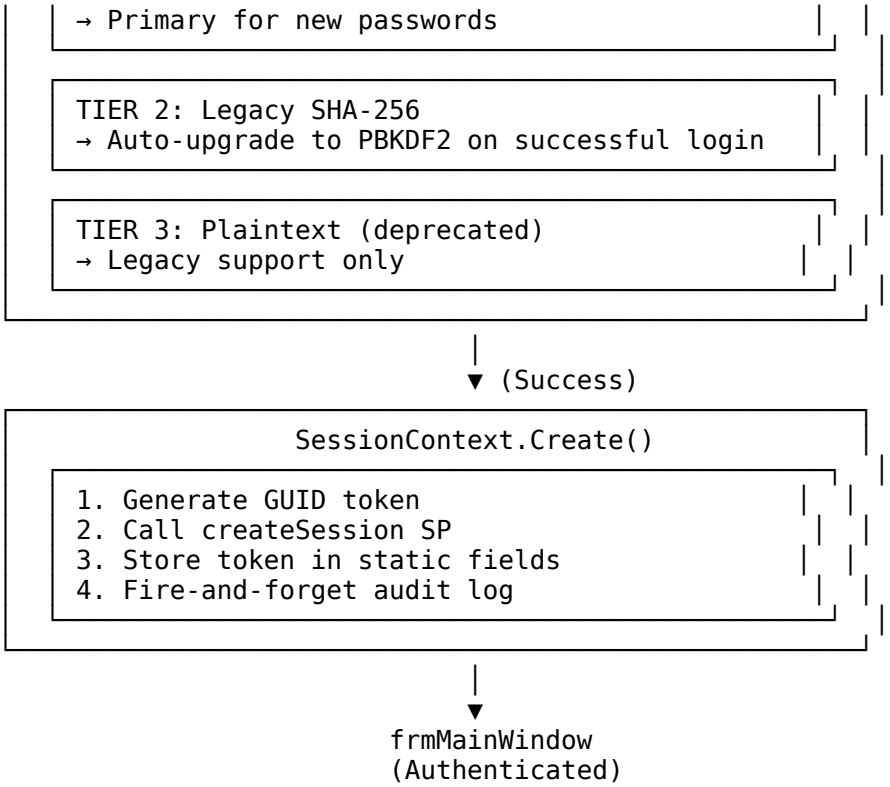
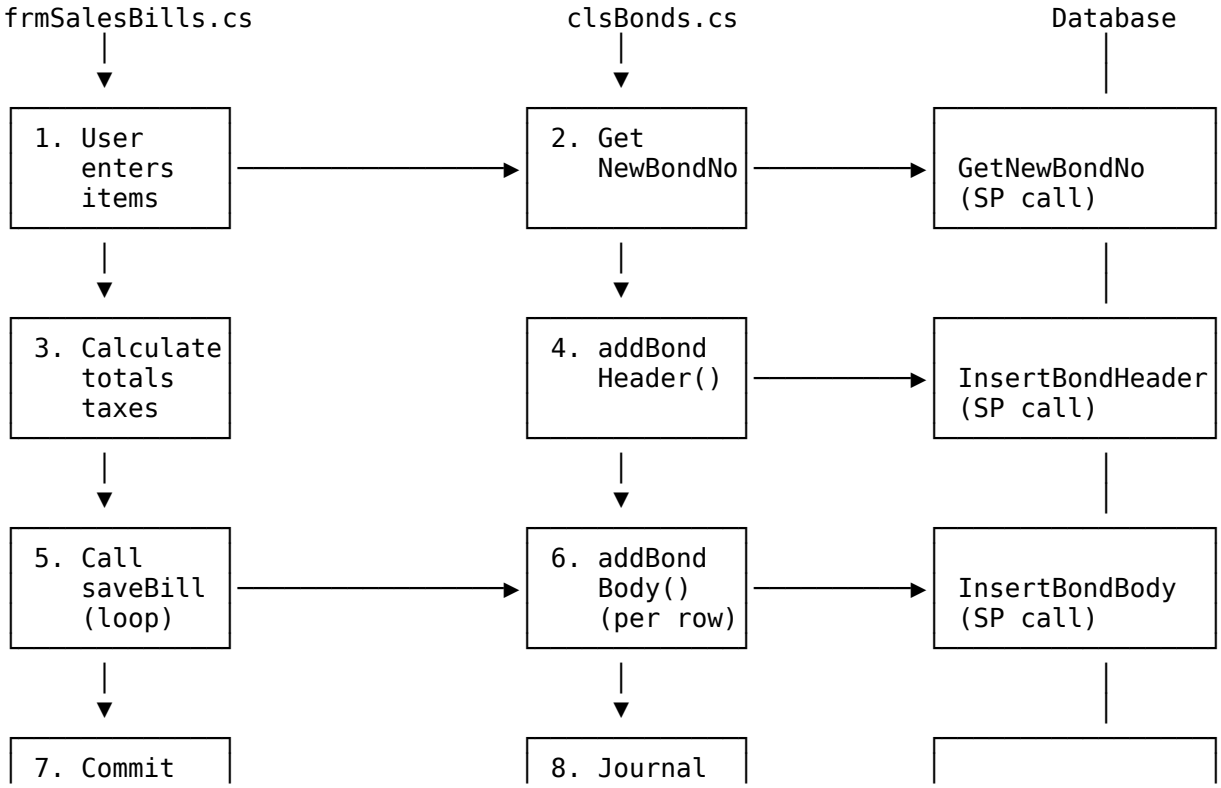
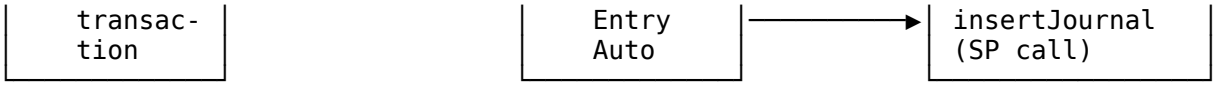
2. تدفق البيانات والعمليات | Data & Process Flows

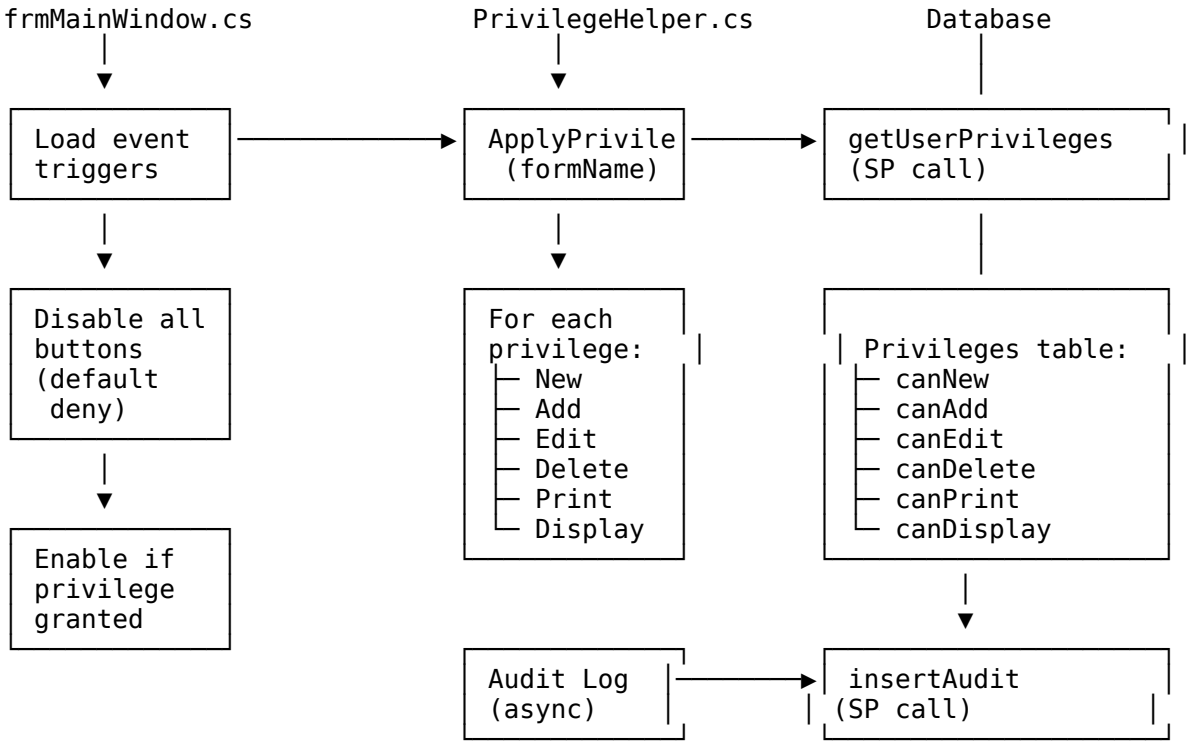
2.1 تدفق المصادقة والتحقق | Authentication Flow

```
┌──────────────┐     ┌──────────────┐     ┌──────────────┐     ┌──────────────┐
│  User Input  │ ──► │  frmLogin    │ ──► │  clsUsers    │ ──► │  Password    │
│  (Creds)     │     │  .Login()    │     │  .Login()    │     │  Helper      │
└──────────────┘     └──────────────┘     └──────┬───────┘     └──────┬───────┘
                                                 │                    │
                                                 ▼                    ▼
                                    ┌──────────────────────────────────────────┐
                                    │        Password Verification (3-Tier)    │
                                    │  ┌────────────────────────────────────┐  │
                                    │  │ TIER 1: PBKDF2-SHA256 (100k iterations)  │
                                    │  │                                    │  │
```




```

### 2.2 تدفق عملية البيع | Sales Bill Flow

```





```

### 2.3 تدفق نظام الصلاحيات | Privilege System Flow

```



```

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## 3. العلاقات والاعتماديات | Relationships & Dependencies

### 3.1 Dependency Graph

```
PL Layer (74 Forms)
    │
    ├──► BL.Security
    │   ├─ PasswordHelper (standalone, no deps)
    │   ├─ AuditHelper (standalone, fire-and-forget)
    │   ├─ SessionContext ──► AuditHelper
    │   └─ PrivilegeHelper ──► AuditHelper
    │
    ├──► BL.Users
    │   └─ clsUsers ──► PasswordHelper, SessionContext, AuditHelper
    │
    ├──► BL.Accounts
    │   └─ clsAccounts ──► AuditHelper
    │
    ├──► BL.Journal
```
```

```
     │      └─ clsjournal ──► AuditHelper
     │
     ├──► BL.Sales
     │      └─ clsSales ──► AuditHelper
     │
     ├──► BL.Bonds
     │      ├─ clsBonds ──► AuditHelper, clsjournal
     │      └─ clsBonds ──► clsAccounts (for validation)
     │
     └──► BL.Stores
            └─ clsInventory ──► AuditHelper, clsAccounts

All BL Classes
     │
     └──► DAL.clsCN (SINGLE ENTRY POINT)
               │
               └──► SQL Server Stored Procedures
```

3.2 Cross-Layer Communication

From Layer	To Layer	Method	Purpose
PL.Forms	BL	`new clsXXX()`	Instantiate business class
PL.Forms	BL	`classInstance.method()`	Call business logic
BL	DAL	`new clsCN()`	Database connection
BL	DAL	`clsCN.GetDataTable()`	Query execution
DAL	DB	SP calls	Data persistence

3.3 Static State Dependencies

```
┌──────────────────────────────────────────────────────────────┐
│                  Application Domain Lifetime                 │
│  ═══════════════════════════════════════════════════════════ │
│                                                              │
│  SessionContext (Static Class)                               │
│  ├─ CurrentUserCode: int                                     │
│  ├─ CurrentUserID: string                                    │
│  ├─ CurrentBraCode: int                                      │
│  ├─ SessionToken: Guid?                                      │
│  └─ LastActivity: DateTime                                   │
│                                                              │
│  [Single instance per application]                           │
│  [Survives across forms]                                     │
│  [Lost on application restart]                               │
└──────────────────────────────────────────────────────────────┘
```

---

4. مسارات التنفيذ وسير العمل | Execution Paths

4.1 Application Startup Path

```
Program.Main()
```

```
    |
    ▼
ApplicationConfiguration.Initialize()
    |
    ▼
Application.Run(new frmLogin())
    |
    ▼
[User sees login screen - app waits for input]
```

4.2 Login Success Path

```
frmLogin.btnLogin_Click()
    |
    |──► txtBranch.Text, txtUser.Text, txtPWD.Text
    |
    ▼
clsUsers.Login(branch, userID, password)
    |
    |──► Query DB for user record
    |
    |──► Verify password (3-tier)
    |
    └──► Return DataTable with user info
         |
         ▼
    (if rows > 0)
         |
         ▼
SessionContext.Create(userCode, userID, braCode)
    |
    |──► new clsCN()
    |
    |──► new SqlParameter[5]
    |
    |──► cmd.Parameters.AddRange(allPara)
    |
    |──► cmd.ExecuteNonQuery()
    |
    |──► SessionContext fields populated
    |
    └──► AuditHelper.LogSessionCreated() [fire-and-forget]
         |
         ▼
frmMainWindow fmw = new frmMainWindow()
    |
    ▼
fmw.Show()
    |
    ▼
this.Hide()
```

4.3 Sales Bill Save Path

```
frmSalesBills.btnSave_Click()
     │
     ├──► Validate inputs
     │
     ├──► Calculate totals
     │
     ▼
clsBonds bonds = new clsBonds()
     │
     ├──► bonds.GetNewBondNo(braCode, bondType)
     │       └──► DAL → GetNewBondNo SP
     │
     ├──► bonds.addBondHeader(...)
     │       └──► DAL → InsertBondHeader SP
     │
     ├──► For each item row:
     │       └──► bonds.addBondBody(...)
     │               └──► DAL → InsertBondBody SP
     │
     ├──► bonds.addJournalHeader(...)
     │       └──► DAL → insertJournalHeader SP
     │
     ├──► For each account affected:
     │       └──► bonds.addJournalBody(...)
     │               └──► DAL → insertJournalBody SP
     │
     └──► AuditHelper.LogBillCreated() [fire-and-forget]
          │
          ▼
MessageBox.Show("تم الحفظ بنجاح")
     │
     ▼
Clear form / Refresh
```

4.4 Session Validation Path (On each form load)

```
frmXXX_Load()
     │
     ▼
if (SessionContext.SessionToken == null)
     │
     ├──► MessageBox.Show("انتهت الجلسة")
     │
     └──► Application.Exit()
          │
          ▼
clsCN.ValidateSession(SessionContext.SessionToken)
     │
     ├──► Call validateSession SP
     │
     ├──► Check result (valid/expired)
     │
     └──► If expired:
           ├──► SessionContext.Clear()
```

```
          ├─► MessageBox.Show("انتهت الجلسة")
          └─► Application.Exit()
```

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5. نقاط القوة والمخاطر والاختناقات | Strengths, Risks, Bottlenecks

5.1 نقاط القوة | Strengths

#	Strength	Description	Impact
1	**PBKDF2-SHA256**	100,000 iterations for password hashing	OWASP 2023 compliant, high security
2	**SQL Injection Prevention**	Regex validation in DAL.ValidateStoredProcedureCall()	Prevents all known SQL injection vectors
3	**Token-based Sessions**	GUID tokens with 1-hour sliding expiry	Secure, stateless, scalable
4	**Async Audit Logging**	Fire-and-forget pattern with Task.Run()	Non-blocking, no performance impact
5	**Privilege System**	6-permission model per screen	Fine-grained access control
6	**Double-entry Accounting**	Journal header/body structure	Financial integrity
7	**Multi-tenant Support**	braCode in all operations	Branch isolation
8	**3-Tier Architecture**	Clean separation PL/BL/DAL	Maintainable, testable

5.2 المخاطر | Risks

#	Risk	Severity	Mitigation
1	**Static Session State**	High	Hard to test, memory leaks possible
2	**Plaintext Password Tier**	Medium	Legacy support only, logs warning
3	**No Encryption in Transit**	High	Add TLS to SQL connection
4	**Session Token in URL**	Medium	Ensure POST-only for sensitive ops
5	**No Rate Limiting**	Medium	Add login attempt throttling
6	**Sync Call for ValidateSession**	Medium	Consider async validation
7	**No Parameterized Queries in SP**	Low	SPs use parameters, safe
8	**Centralized Exception Handling**	Low	Add global exception handler

5.3 الاختناقات التقنية | Bottlenecks

#	Bottleneck	Location	Type	Recommendation
1	**DAL Connection per call**	clsCN	Performance	Implement connection pooling
2	**Sync validation on each form**	SessionContext	Latency	Cache validation, async refresh
3	**No lazy loading**	All forms	Memory	Implement virtual scrolling for grids
4	**Large DataTable returns**	getListOfAccounts, getAllCustomers	Memory	Implement pagination
5	**No caching layer**	All BL classes	Performance	Add Redis/MemoryCache

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6. فرص التحسين | Improvement Opportunities

6.1 Critical Improvements (Priority 1)

```
┌──────────────────────────────────────────────────────────────────────────────┐
│ 1. Add Connection Pooling to clsCN                                           │
│ ═══════════════════════════════════════════════════════════════════════════ │
│                                                                              │
│ Current:                                                                     │
│   private SqlConnection GetConnection()                                      │
│   {                                                                          │
│       SqlConnection cn = new SqlConnection(connectionString);                │
│       cn.Open();  // New connection every time                               │
│       return cn;                                                             │
│   }                                                                          │
│                                                                              │
│ Improved:                                                                    │
│   private static readonly object _lock = new object();                       │
│   private static SqlConnection _cachedConnection;                            │
│   private static DateTime _lastConnectionTime;                               │
│   private const int ConnectionTimeoutMinutes = 5;                            │
│                                                                              │
│   private SqlConnection GetConnection()                                      │
│   {                                                                          │
│       lock (_lock)                                                           │
│       {                                                                      │
│           if (_cachedConnection != null &&                                   │
│               _cachedConnection.State == ConnectionState.Open &&             │
│               (DateTime.Now - _lastConnectionTime).TotalMinutes <            │
ConnectionTimeoutMinutes)                                                      │
│               return _cachedConnection;                                      │
│                                                                              │
│           _cachedConnection?.Dispose();                                      │
│           _cachedConnection = new SqlConnection(connectionString);           │
│           _cachedConnection.Open();                                          │
│           _lastConnectionTime = DateTime.Now;                                │
│           return _cachedConnection;                                          │
│       }                                                                      │
│   }                                                                          │
```

2. Add Global Exception Handler

In Program.cs:

```
Application.ThreadException += (s, e) =>
{
    AuditHelper.LogException(e.Exception);
    MessageBox.Show($"حدث خطأ غير متوقع: {e.Exception.Message}",
        "خطأ", MessageBoxButtons.OK, MessageBoxIcon.Error);
};

AppDomain.CurrentDomain.UnhandledException += (s, e) =>
{
    var ex = e.ExceptionObject as Exception;
    AuditHelper.LogException(ex);
};
```

3. Add Login Rate Limiting

In Database:

```
CREATE TABLE LoginAttempts (
    AttemptID INT IDENTITY PRIMARY KEY,
    UserID NVARCHAR(15),
    IPAddress NVARCHAR(50),
    AttemptTime DATETIME,
    Success BIT
);

In clsUsers.Login():
var attempts= cn.GetDataTable("checkRecentAttempts",
```

```
    new SqlParameter("@userID", userID),
    new SqlParameter("@windowMinutes", 15));

if (attempts.Rows.Count >= 5)
    throw new AccountLockedException("تم تجاوز محاولات تسجيل الدخول");
```

6.2 High Priority Improvements (Priority 2)

```
4. Implement Async Session Validation
═══════════════════════════════════════

public static async Task<bool> ValidateSessionAsync()
{
    if (SessionToken == null) return false;

    return await Task.Run(() => {
        try {
            clsCN cn = new clsCN();
            var result = cn.GetDataTable("validateSession",
                new SqlParameter("@sessionToken", SessionToken.Value));
            return result.Rows.Count > 0;
        }
        catch { return false; }
    });
}

Usage:
private async void frmXXX_Load(object sender, EventArgs e)
{
    bool isValid = await SessionContext.ValidateSessionAsync();
    if (!isValid) { /* redirect to login */ }
}
```

```
5. Add Data Pagination for Large Result Sets
```

```
═══════════════════════════════════════════════════════════════════════════

In clsAccounts:


public DataTable getAccountsPage(int braCode, int page, int pageSize)
{

    clsCN cn = new clsCN();
    return cn.GetDataTable("getAccountsPaginated",
        new SqlParameter("@braCode", braCode),

        new SqlParameter("@page", page),

        new SqlParameter("@pageSize", pageSize));
}


In Stored Procedure:

CREATE PROCEDURE getAccountsPaginated

    @braCode INT, @page INT, @pageSize INT

AS
BEGIN
    SELECT * FROM ChartOfAccounts

    WHERE braCode = @braCode
    ORDER BY accCode
    OFFSET (@page - 1) * @pageSize ROWS
    FETCH NEXT @pageSize ROWS ONLY;
END
```

```
6. Add TLS Encryption for Database Connection

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In clsCN constructor:


SqlConnectionStringBuilder builder = new SqlConnectionStringBuilder(
    connectionString);

builder.Encrypt = true;
builder.TrustServerCertificate = false;  // Use valid cert in production
builder.ConnectTimeout = 30;



Or in connection string:
```

```
"Server=...;Database=...;Encrypt=true;TrustServerCertificate=false;..."
```

6.3 Medium Priority Improvements (Priority 3)

```
7. Add Password Complexity Validation
═══════════════════════════════════════════════════════════════════════

In PasswordHelper:

public static ValidationResult ValidateStrength(string password)
{
    if (password.Length < 8)
        return new ValidationResult(false, "8 أحرف على الأقل");

    if (!password.Any(char.IsUpper))
        return new ValidationResult(false, "حرف كبير واحد على الأقل");

    if (!password.Any(char.IsLower))
        return new ValidationResult(false, "حرف صغير واحد على الأقل");

    if (!password.Any(char.IsDigit))
        return new ValidationResult(false, "رقم واحد على الأقل");

    if (!password.Any(c => !char.IsLetterOrDigit(c)))
        return new ValidationResult(false, "رمز خاص واحد على الأقل");

    return new ValidationResult(true, "قوة كلمة المرور: قوية");
}
```

```
8. Add Audit Log Retention Policy
═══════════════════════════════════════════════════════════════════════

Create scheduled job (SQL Agent):

CREATE PROCEDURE cleanupOldAuditLogs
AS
```

```
BEGIN
    DECLARE @cutoffDate DATETIME = DATEADD(YEAR, -1, GETDATE());

    DELETE FROM AuditLog WHERE AuditTime < @cutoffDate;
END



Schedule: Daily at 2:00 AM
```

```
9. Implement Business Layer Interfaces for Testability
═══════════════════════════════════════════════════════


public interface IUserAuthentication

{

    DataTable Login(int braCode, string userID, string password);
    void ChangePassword(int userCode, string oldPwd, string newPwd);
    bool ValidateSession(Guid token);
}


public class clsUsers : IUserAuthentication { ... }



public interface IAuditLogger
{

    void Log(string eventType, string actionName, ...);
    void LogException(Exception ex);
}


public class AuditHelper : IAuditLogger { ... }
```

```

---

## 7. ملخص المشروع | Project Summary

### 7.1 Statistics

| Metric | Count |
|--------|-------|
| Total Forms (PL) | 74 |
| Business Classes (BL) | 13 |
| DAL Classes | 1 |
| Stored Procedures | 50+ |
| Database Tables | 40+ |
```

| Lines of Code (approx) | 15,000+ |

7.2 Security Posture

Aspect	Status	Notes
Password Storage	✅ Strong	PBKDF2-SHA256, 100k iterations
SQL Injection	✅ Protected	Regex validation in DAL
Session Management	✅ Secure	GUID tokens, sliding expiry
Access Control	✅ Implemented	6-permission per screen
Audit Logging	✅ Active	Fire-and-forget async
Transport Security	⚠ Needs Work	Add TLS to DB connection
Rate Limiting	❌ Missing	Add login attempt throttling

7.3 Architecture Compliance

Principle	Status	Notes
3-Tier Separation	✅ Full	PL→BL→DAL→DB
No Business Logic in PL	✅ Compliant	All logic in BL
DAL as Single Entry	✅ Compliant	One clsCN class
Stored Procedure Usage	✅ Compliant	No inline SQL
Transaction Support	✅ Implemented	In clsCN

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8. خارطة طريق التنفيذ | Implementation Roadmap

```
Phase 1: Security Hardening (Week 1-2)
├── Add TLS encryption to DB connection
├── Implement login rate limiting
├── Add global exception handler
└── Password complexity validation

Phase 2: Performance Optimization (Week 3-4)
├── Implement connection pooling
├── Add async session validation
└── Pagination for large result sets

Phase 3: Maintainability (Week 5-6)
├── Add BL interfaces for testing
├── Implement audit log retention
└── Document all SPs

Phase 4: Monitoring (Week 7-8)
├── Add performance counters
├── Implement health check endpoint
└── Dashboard for key metrics
```

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Architecture: WinForms 3-Tier (PL/BL/DAL)
Version: 1.0